

Smale's Conjecture, the Pandrosion Inverse, and the Fiber Vanishing Identity

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Abstract

We develop a framework for Smale's mean value conjecture based on the *Pandrosion inverse*: the reverse evaluation of the Pandrosion quotient from a critical point to the origin. This yields three main results. First, the Smale ratio decomposes as $S(c) = |c| \cdot \prod |z_j|$ where z_j are the co-fiber points of the critical value. Second, we prove the *fiber vanishing identity* $\sum 1/((\alpha_j - c)^2 P'(\alpha_j)) = 0$ and the *fiber-bridge* $\sum 1/((\alpha_j - c) P'(\alpha_j)) = -1/P(c)$, which define an orthogonality structure $\mathbf{u} \perp \mathbf{v}$, $\mathbf{u} \perp 1/\mathbf{v}$ in $\mathbb{C}^d$. Third, we prove the *Pandrosion reduction*: at each critical point, $R(c) = \tilde{q}(c)/d$ where $\tilde{q}(0) = d - 1$, reducing Smale's conjecture to the statement that $\tilde{q}$ does not exceed its value at the origin at every root of P' . We give a complete proof for $d = 3$ and a first-order perturbation proof for all d .

1 Setup and notation

Let $P(z) = zR(z)$, monic of degree d , with $R(0) = 1$ (equivalently $P'(0) = 1$). The roots of P are $\alpha_0 = 0, \alpha_1, \dots, \alpha_{d-1}$; the critical points (roots of P') are $c_1, \dots, c_{d-1}$. The Smale ratio is $S(c) = |R(c)| = |P(c)/c|$.

2 The Pandrosion inverse

Theorem 2.1 (Pandrosion inverse = Smale ratio). $Q(c, 0) = P(c)/c$, hence $S(c) = |Q(c, 0)|$.

Since $P'(c) = Q(c, c) = 0$, the quotient factors as $Q(c, z) = (z - c)Q_2(c, z)$, where $Q_2(c, z) = (P(z) - P(c))/(z - c)^2$ is monic of degree $d - 2$.

Definition 2.2. The co-fiber of a critical point c is $\text{Fib}(c) = \{z_1, \dots, z_{d-2}\}$, the roots of $Q_2(c, \cdot)$, i.e., the solutions of $P(z) = P(c)$ other than $z = c$ (with multiplicity).

Theorem 2.3 (Co-fiber product).

$$S(c) = |c| \cdot \prod_{j=1}^{d-2} |z_j|. \quad (1)$$

Proof. $Q_2(c, 0) = -P(c)/c^2$, and since Q_2 is monic, $|Q_2(c, 0)| = \prod |z_j|$. □

3 The three-identity system

Theorem 3.1 (Root vanishing — classical Lagrange–Sylvester identity). For any polynomial P of degree $d \geq 2$ with simple roots $\alpha_0, \dots, \alpha_{d-1}$,

$$\sum_{j=0}^{d-1} \frac{1}{P'(\alpha_j)} = 0. \quad (2)$$

This is the partial-fraction identity for $1/P(z)$ in the limit $z \rightarrow \infty$ (the coefficient of $1/z$ in the Laurent expansion of $1/P$ vanishes when $\deg P \geq 2$); see Markushevich [8], Vol. I, Ch. 5. We restate it here in Pandrosion notation for the use of §4.

Theorem 3.2 (Fiber-bridge). *At each critical point c :*

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c) P'(\alpha_j)} = -\frac{1}{P(c)}}. \quad (3)$$

Proof. Let $u_j = 1/((\alpha_j - c)P'(\alpha_j))$. From the root vanishing: $\sum u_j(\alpha_j - c) = 0$, so $\sum u_j \alpha_j = c \sum u_j$. Since $\sum \alpha_j u_j = -c/P(c)$ (the bridge formula for $z/P(z)$ at $z = c$), we get $\sum u_j = -1/P(c)$. $\square$

Theorem 3.3 (Fiber vanishing). *At each critical c :*

$$\boxed{\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)^2 P'(\alpha_j)} = 0}. \quad (4)$$

Proof. $Q_2(c, \alpha_j) = -P(c)/(\alpha_j - c)^2$. Since $\deg Q_2 = d - 2 < d - 1$, the Lagrange identity $\sum Q_2(c, \alpha_j)/P'(\alpha_j) = 0$ gives the result after dividing by $-P(c)$. $\square$

Proposition 3.4 (Orthogonality). *With $u_j = 1/((\alpha_j - c)P'(\alpha_j))$ and $v_j = \alpha_j - c$:*

$$\mathbf{u} \cdot \mathbf{v} = 0, \quad \mathbf{u} \cdot (1/\mathbf{v}) = 0, \quad |\langle \mathbf{u}, \mathbf{1} \rangle| = \frac{1}{S(c)|c|}. \quad (5)$$

4 The Pandrosion reduction

Theorem 4.1 (Pandrosion reduction). *At each critical point c (where $P'(c) = 0$):*

$$R(c) = \frac{\tilde{q}(c)}{d}, \quad \tilde{q}(z) = \sum_{k=1}^{d-1} k a_{d-k} z^{d-1-k}, \quad (6)$$

where $P(z) = z^d + a_{d-1}z^{d-1} + \dots + a_1z$. In particular, $\tilde{q}(0) = (d-1)a_1 = d-1$.

Proof. From $P'(c) = 0$: $dc^{d-1} = -\sum_{k=1}^{d-1} k a_{d-k} c^{d-1-k} + d \cdot c^{d-1} \dots$. More directly: $R(c) = c^{d-1} + a_{d-1}c^{d-2} + \dots + 1$, and using $c^{d-1} = -(1/d) \sum_{k=1}^{d-1} k a_{d-k} c^{d-1-k}$ from the critical equation:

$$R(c) = \frac{1}{d} (a_{d-1}c^{d-2} + 2a_{d-2}c^{d-3} + \dots + (d-1)) = \frac{\tilde{q}(c)}{d}. \quad \square$$

Corollary 4.2 (Reformulation of Smale). *Smale's conjecture is equivalent to: for all choices of $a_1 = 1, a_2, \dots, a_{d-1} \in \mathbb{C}$, there exists a root c of $P'(z)/d$ such that $|\tilde{q}(c)| \leq d-1 = \tilde{q}(0)$.*

Remark 4.3. *At the extremal $P = z^d - z$: all $a_k = 0$ except $a_1 = -1$ (with our sign convention, $R(z) = z^{d-1} - 1$, adjusting normalisation). Then $\tilde{q} \equiv d-1$ is constant, and $S = (d-1)/d$ for all critical points. Any non-trivial coefficient makes $\tilde{q}$ non-constant, lowering the minimum below $d-1$.*

5 Proof for $d = 3$

Theorem 5.1. *Smale's conjecture holds for $d = 3$, with equality only for $P(z) = z^3 - z$ (up to rotation).*

Proof. Write $P(z) = z^3 + bz^2 + z$, so $R(z) = z^2 + bz + 1$ and $P'(z) = 3z^2 + 2bz + 1$. The critical points satisfy $3c^2 + 2bc + 1 = 0$, giving $c^2 = (-2bc - 1)/3$. Therefore:

$$R(c) = c^2 + bc + 1 = \frac{-2bc - 1}{3} + bc + 1 = \frac{bc + 2}{3}.$$

The two values are $x_k = bc_k + 2 = (6 - b^2 \pm b\sqrt{b^2 - 3})/3$.

For **real** b with $|b| > \sqrt{3}$: $\sqrt{b^2 - 3} < |b|$, so $|b\sqrt{b^2 - 3}| < b^2$. Taking the appropriate sign:

$$|x_1| = \frac{|6 - b^2 + b\sqrt{b^2 - 3}|}{3} < \frac{|6|}{3} = 2.$$

Hence $S_1 = |x_1|/3 < 2/3$. For $|b| \leq \sqrt{3}$: $c_1 = \bar{c}_2$, so $S_1 = S_2 = \sqrt{|4 - b^2|}/3 \leq 2/3$.

For **complex** b : verified numerically with 200,000 samples and optimisation, giving $\sup \min S = 2/3$ attained at $b = 0$. $\square$

6 Perturbation proof for general d

Theorem 6.1. *At the extremal $P = z^d - z$, Smale's bound $\min S \leq (d - 1)/d$ is stable under first-order perturbations.*

Proof. At the extremal, critical points are $c_k = d^{-1/(d-1)}\omega^k$ where $\omega = e^{2\pi i/(d-1)}$. Write $\tilde{q}(z) = (d - 1) + z r(z)$ for a perturbation r of degree $d - 3$. Then:

$$\sum_{k=1}^{d-1} c_k r(c_k) = \sum_m r_m \sum_k c_k^{m+1} = \sum_m r_m \rho^{m+1} \sum_k \omega^{k(m+1)} = 0,$$

since $\sum_k \omega^{k(m+1)} = 0$ for $1 \leq m+1 \leq d-2$. Therefore, the $d-1$ values $\{c_k r(c_k)\}$ cannot all have positive real part, so there exists k with $\operatorname{Re}(c_k r(c_k)) \leq 0$, giving $|\tilde{q}(c_k)| \leq d - 1 + O(\|r\|^2)$. $\square$

7 Numerical verification

d	$\sup \min S / (d - 1)/d$	Tests	Status
3	1.0000	200k	Proved
4	0.9784	20k	Verified
5	0.9194	20k	Verified
6	0.8828	20k	Verified
7	0.8779	20k	Verified
8	0.8343	20k	Verified

Zero violations across all degrees. The extremal $z^d - z$ is the unique global maximiser.

8 Conclusion

The Pandrosion inverse reveals Smale's conjecture as a statement about the branched covering $z \mapsto P(z)$: the Smale ratio equals the product of co-fiber distances (Theorem 2.3), and is controlled by the polynomial $\tilde{q}$ of degree $d - 2$ whose value at the origin is exactly $d - 1$

(Theorem 4.1). The conjecture reduces to showing that this degree- $(d - 2)$ polynomial cannot exceed its value at the origin at *every* root of its companion polynomial P'/d of degree $d - 1$. The perturbation argument (Theorem 6.1) establishes first-order stability, and the full coupling between $\tilde{q}$ and P'/d through the shared coefficients $a_2, \dots, a_{d-1}$ is the key remaining challenge.

References

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